

3175. Find The First Player to win K Games in a Row

Medium Topics  Hint

A competition consists of n players numbered from 0 to $n - 1$.

You are given an integer array `skills` of size n and a **positive** integer k , where `skills[i]` is the skill level of player i . All integers in `skills` are **unique**.

All players are standing in a queue in order from player 0 to player $n - 1$.

The competition process is as follows:

- The first two players in the queue play a game, and the player with the **higher** skill level wins.
- After the game, the winner stays at the beginning of the queue, and the loser goes to the end of it.

The winner of the competition is the **first** player who wins k games **in a row**.

Return the initial index of the *winning* player.

Example 1:

Input: `skills = [4,2,6,3,9]`, `k = 2`

Output: `2`

Explanation:

Initially, the queue of players is `[0,1,2,3,4]`. The following process happens:

- Players 0 and 1 play a game, since the skill of player 0 is higher than that of player 1, player 0 wins. The resulting queue is `[0,2,3,4,1]`.
- Players 0 and 2 play a game, since the skill of player 2 is higher than that of player 0, player 2 wins. The resulting queue is `[2,3,4,1,0]`.
- Players 2 and 3 play a game, since the skill of player 2 is higher than that of player 3, player 2 wins. The resulting queue is `[2,4,1,0,3]`.

Player 2 won `k = 2` games in a row, so the winner is player 2.

Example 2:

Input: `skills = [2,5,4]`, `k = 3`

Output: `1`

Explanation:

Initially, the queue of players is `[0,1,2]`. The following process happens:

- Players 0 and 1 play a game, since the skill of player 1 is higher than that of player 0, player 1 wins. The resulting queue is `[1,2,0]`.
- Players 1 and 2 play a game, since the skill of player 1 is higher than that of player 2, player 1 wins. The resulting queue is `[1,0,2]`.
- Players 1 and 0 play a game, since the skill of player 1 is higher than that of player 0, player 1 wins. The resulting queue is `[1,2,0]`.

Player 1 won `k = 3` games in a row, so the winner is player 1.

Constraints:

- `n == skills.length`
- `2 <= n <= 105`
- `1 <= k <= 109`
- `1 <= skills[i] <= 106`
- All integers in `skills` are unique.

Seen this question in a real interview before? 1/6

Yes No

Accepted 36,771/91.4K | Acceptance Rate 40.2%

Topics



Hint 1



Hint 2



Discussion (42)



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